

Day 1 — Complete Toolkit & Notes

Aaj jo kuch seekha — har tool, uska kaam, format aur example — sab ek jagah.

Jab bhi kuch bhoolo, ye khol lo. Easy Hinglish, koi complexity nahi. 🌙

25+ tools

12 Bandit levels

1 din 🔥

01

⚙️ Setup & SSH

Safe Linux lab (WSL/Ubuntu) aur remote login.

```
wsl --install
```

Windows ke andar Linux (Ubuntu) install karta hai. Admin PowerShell me chalao

```
ssh user@host -p port
```

Kisi doosre computer par door se (remote) login karta hai. SSH = Secure Shell, port 22

Example `ssh bandit0@bandit.labs.overthewire.org -p 2220`

```
exit
```

Remote session se bahar niklo (wapas apni machine par).



Golden Rule: Hacking hamesha VM / online lab me. Sirf apne ya likhit-permission wale systems par. Windows data hamesha safe rahega.

02



Navigation

Folders ke andar-bahar jaana aur dekhna.

```
pwd
```

Abhi kis folder me ho, wo dikhata hai. `Print Working Directory`

```
ls
```

Folder ke andar kya-kya hai, list karta hai (hidden nahi).

```
ls -a
```

Hidden files bhi dikhao (jo `.` se shuru hoti hain). `a = all`

```
ls -la
```

Hidden + पूरी detail (permission, size, date) dono. `l = long`

```
cd folder
```

Us folder ke andar jao. `Change Directory` - sirf folder me, file me nahi!

```
cd ..
```

Ek folder peeche/bahar jao. space zaroori: `cd ..` (`cd..` galat)

```
cd ~
```

Seedha apne home folder me. `~ = /home/username`

```
cd /
```

Sabse upar (root) folder me — Linux ki jad.

clear

Screen saaf karo. Ctrl + L bhi

03



File & Folder

Banana, padhna, copy, rename, delete.

```
mkdir naam
```

Naya folder banao. `Make Directory`

Example `mkdir hacking-lab`

```
touch file.txt
```

Nayi khaali file banao.

```
echo "text" > file
```

File me text daalo (purana mit jayega). `ek > = overwrite`

Example `echo "hello" > notes.txt`

```
echo "text" >> file
```

Purane ke neeche jodo (kuch mitta nahi). `do >> = append`

```
cat file
```

File ke andar ka text padho. `conCATenate`

```
cp file copy
```

File ki copy banao. `CoPy`

Example `cp notes.txt backup.txt`

```
mv old new
```

Naam badlo ya move karo. `MoVe` • maujood file par karoge to overwrite ho jayegi!

```
rm file
```

File delete karo. ReMove • permanent, koi Recycle Bin nahi!

```
rm -r folder
```

Poora folder (andar sab kuch) delete karo. `r` = recursive

Example `rm -r /tmp/mera_folder`

04



System & Install

Admin power aur software install (Ubuntu ka "App Store").

```
sudo command
```

Admin (root) power ke saath chalao — password maangega. Super User DO

```
sudo apt update
```

Package list refresh — naya kya available hai, pata karo.

```
sudo apt install naam -y
```

Software install karo. `-y` = "haan, kar do"

Example `sudo apt install nmap -y`

```
tool --version
```

Kisi tool ka version check karo (install hua ya nahi).



Password type karte waqt screen par kuch nahi dikhega — ye normal hai (security). Bas type karke Enter.

05



Nmap — Scanning

Target ke khule "ports" (darwaze) aur unke software dhoondhna.

```
nmap target
```

Target ke ports scan karo (naam ya IP). Kaun sa darwaza khula hai

Example `nmap scanme.nmap.org`

```
nmap -sV target
```

Har khule port par kaun sa software + version chal raha hai. *Service Version*

```
nmap localhost
```

Apna hi computer scan karo (127.0.0.1) — practice ke liye.

Port ki 3 haalat (STATE):

● **open**

Darwaza khula — service chal rahi. Possible entry point.

● **closed**

Darwaza band hai (par wahan hai).

● **filtered**

Firewall beech me — pata nahi khula/band. Achhi security.

06



Recon Soch — Vulnerability dhoondhna

Version mila? Ab uski kamzori kaise dhoondhte hain.

1

`nmap -sV` se software + **version** nikaalo

2

exploit-db.com / Google par us version ki **CVE / known bug** dhoondo

3

Range check: version sach me kamzor range me hai? (jaise `< 6.6` = 6.6 se neeche hi)

4

Faisla: raasta khula hai ya band. Har port par dohrao.



Bada lesson: sirf software ka naam match hona kaafi nahi — exact **version** bhi kamzor range me hona chahiye.

07



Dhoondhna & Filter

Bahut saari files/text me se kaam ki cheez nikaalna.

```
file naam
```

File ke andar kis type ka data hai batao (text/binary/gzip...). naam nahi, andar dekh kar

Example file data.txt

```
./*
```

* = "sab kuch" (wildcard). file ./* = saari files ka type ek saath.

```
find . -type f -size 1033c
```

Properties se file dhoondo (saare sub-folders me). f=file, c=bytes

```
find / -user U -group G -size 33c 2>/dev/null
```

Poore server (/) me owner/group/size se dhoondo. 2>/dev/null = errors chhupao

```
grep "word" file
```

File ke andar sirf woh line jisme word ho. "word" ki jagah asli word daalo!

Example grep millionth data.txt

-type f	sirf files (folders nahi)
-size 33c	exactly 33 bytes (c = bytes)
-user / -group	owner user / group ke hisaab se

08



Pipe & Jodna

Ek command ka output doosre ko dena — Linux ka superpower.

```
command1 | command2
```

Pipe | — command1 ka OUTPUT command2 ko INPUT ki tarah do. Keyboard: Shift + \

Example `cat data.txt | grep word`

```
sort file
```

Lines ko order (alphabetically) me lagao — same lines paas aa jaati hain.

```
sort file | uniq -u
```

Sirf woh line jo **akeli (unique)** ho. `uniq -u`; pehle `sort` zaroori



Pipe kab NAHI: jab commands alag-alag kaam karein (jaise `mv`, `gunzip`, `file`) — unhe **alag lines** me chalo. Pipe sirf "output aage bhejna hai" tab.

09



Chhupi / Ajeeb Files

Hidden files aur weird naam wali files handle karna.

```
ls -a
```

Hidden files dikhao (jo `.` se shuru — jaise `.bashrc`).

```
cat ./-file
```

`./` = "yahin, is folder me". Naam jab `-` ya `.` se shuru ho tab lagao. `-` ko `cat` option samajh leta hai

Example `cat ./- · cat ./...Hiding`

```
cat "spaces in name"
```

Naam me spaces ho to `" "` me daalo. Ya `Tab` key se auto-complete

```
strings file
```

Binary/garbage file me se sirf padhne-layak text nikaalo.

Example `strings data.txt | grep "==="`



Tab key = auto-complete. Aadha naam type karke `Tab` dabao → terminal khud poora kar deta hai. Time bachta hai!

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Decode & Cipher

Chhupe/coded data ko wapas readable banana.

```
base64 -d file
```

Base64 encoded data ko decode karo. -d = decode; == par khatam hone wala text = base64

```
cat file | tr 'A-Za-z' 'N-ZA-Mn-za-m'
```

ROT13 cipher decode (har letter 13 aage shift). tr = translate/replace characters



Encoding ≠ Encryption! Base64/ROT13 = sirf "format/code" — koi bhi ulta karke padh sakta hai, ye **protection nahi**. Encryption = asli tala jise bina key nahi khol sakte.

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Reverse & Decompress

Hexdump ulta karna + baar-baar compress ki hui file kholna (Level 12).

```
xxd -r file > newfile
```

Hexdump ko ulta karke asli (binary) file banao. -r = reverse; > se save

Compression kholne ki table:

gzip	mv X X.gz → gunzip X.gz
bzip2	mv X X.bz2 → bunzip2 X.bz2
POSIX tar	mv X X.tar → tar xf X.tar (naya file → ls)
ASCII text	🚩 cat X → password mil gaya!



Loop: `file X` → table dekho → kholo → phir `file X` ... jab tak "ASCII text"



2 zaroori aadatein: (1) Hamesha **sabse naye** nikle file par kaam karo, purane dabbe chhod do.
(2) Ye steps **alag-alag lines** me — pipe | bilkul nahi.

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Galtiyaan jinse bachna hai

Transcript paste	Poora terminal text (prompt+output) paste mat karo — sirf command.
rm permanent	rm / rm -r se delete = wapas nahi aata. Dekh kar delete karo.
Placeholder	Format me "word", filename = example. Apni asli cheez se replace karo.
cd on file	File me cd nahi hota ("Not a directory") — file padhne ko cat.
Pipe galat	Alag kaam wale commands ko se mat jodo — race/error.
Location	Folder wahin milega jahan wo hai — pwd se check karo "main kahan hun".

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Ethical Rules (yaad rakho)

Permission	Sirf apne / likhit-permission wale / CTF-lab systems par.
No spoilers	Kisi level ka password/solution kahin public post mat karo.
Don't break	Server ko crash/attack mat karo — sirf game/lab jaise design hai.
Clean up	Apni temporary files (jaise /tmp me) baad me delete karo.



Ye game ke rules = asli duniya ke ethical-hacking rules ka chhota version. Bug bounty me bhi: scope ke andar raho, todo mat, sensitive info public mat karo.

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⚡ Quick Tool Index (sab ek nazar me)

pwd — kahan hun

ls / **-a** / **-la** — list / hidden / detail

cd — folder me jao

mkdir — folder banao

touch — file banao

echo > / >> — likho / jodo

cat — file padho

cp / **mv** / **rm** — copy / rename / delete

rm -r — folder delete

sudo — admin power

apt — software install

ssh — remote login

nmap / **-sv** — port / version scan

file — type pehchano

***** — wildcard "sab"

find — properties se dhoondo

grep — text dhoondo

| (pipe) — output aage bhejo

sort / **uniq -u** — order / unique

strings — binary se text

base64 -d — decode

tr — characters badlo (ROT13)

xxd -r — hexdump ulta

gunzip / **bunzip2** — decompress

tar xf — archive kholo

clear / **Tab** — saaf / auto-complete